



Yakeen Plus (2025)

Practice Test- 05

DURATION :200 Minutes

DATE : 16/10/2024

M.MARKS :720

ANSWER KEY

PHYSICS

1. (4)
2. (1)
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CHEMISTRY

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ZOOLOGY

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199. (1)
200. (4)

SECTION-(I) PHYSICS

1. (4)

Radiation $\propto T^4$

$$\frac{E_1}{E_2} = \frac{T_1^4}{T_2^4}$$

$$\frac{E}{E_2} = \frac{T^4 \times (4)^4}{T^4}$$

$$\Rightarrow E_2 = \frac{E}{256}$$

[NEW NCERT Page No. 396]

2. (1)

$$\rho_2 = \rho_1(1 - \gamma \Delta T)$$

$$\frac{-\rho_2 + \rho_1}{\rho_1} = 49 \times 10^{-5} \times 30$$

$$\Rightarrow \frac{\Delta \rho}{\rho_1} = 1.47 \times 10^{-2}$$

[NEW NCERT Page No. 282]

3. (2)

Since U depends upon change in temperature, for isothermal process dT will be zero, So dU will be zero.

So according to first law of thermodynamics

$$dQ = dU + dW$$

$$\rightarrow dQ = dW$$

Since it is expansion so W will be positive and hence Q will be positive

dQ = positive, dU = zero, dW = positive

$$\therefore dQ = dU + dW$$

[NEW NCERT Page No. 308]

4. (3)

Heat is a energy which flows due to temperatures difference.

[NEW NCERT Page No. 280]

5. (1)

$$T_1 = T_2, \text{ hence } \Delta T = 0 \Rightarrow \Delta U = 0$$

[NEW NCERT Page No. 305]

6. (2)

Work done = area under P-V curve

$$W = \pi(-2P_0)(2V_0) = -4\pi P_0 V_0$$

[NEW NCERT Page No. 306]

7. (2)

1st process is isothermal expansion

$$PV = P_2(3V)$$

$$\Rightarrow P_2 = \frac{P}{3}$$

2nd process is adiabatic compression

$$P_2 V_2^\gamma = P_3 V_3^\gamma$$

$$\frac{P}{3}(3V)^\gamma = P_3 V^\gamma$$

$$P_3 = 3^{\gamma-1} P = 3^{\frac{2}{3}} P$$

[NEW NCERT Page No. 313]

8. (4)

In adiabatic process, $\Delta Q = 0$

In adiabatic process

$$\Delta W = \frac{nR}{1-\gamma} [T_2 - T_1]$$

$$\Delta W = -\Delta U$$

[NEW NCERT Page No. 312]

9. (4)

$$dQ = dU + dW$$

In isobaric process

$$dQ = nC_p dT$$

$$dU = nC_v dT$$

$$dW = n(C_p - C_v) dT$$

$$dQ : dU : dW = C_p : C_v : (C_p - C_v)$$

$$= \frac{7}{2} R : \frac{5}{2} R : R$$

$$= 7 : 5 : 2$$

[NEW NCERT Page No. 313]

10. (3)

Temperature determines the direction of flow of heat when two bodies are placed in thermal contact.

[NEW NCERT Page No. 314]

11. (3)

Slope of adiabatic process of P-V diagram

$$\left(\frac{dP}{dV}\right) = \gamma \frac{P}{V} \text{ i.e slope from graph}$$

$$(\text{slope})_2 > (\text{slope})_1$$

$$\Rightarrow \gamma_2 > \gamma_1$$

[NEW NCERT Page No. 305]

12. (2)

[NEW NCERT Page No. 316]

13. (2)

Process is isobaric.

$$C = C_p = C_v + R$$

$$= \frac{3}{2} R + R$$

$$C_p = \frac{5}{2} R$$

[NEW NCERT Page No. 334]

14. (4)
 In process ABC
 $\therefore \Delta Q = \Delta U + \Delta W$
 So $\Delta U = \Delta Q - \Delta W$
 $\Delta U = (400 + 100) - (6 \times 10^4 \times 2 \times 10^{-3})$
 $\Delta U = 500 - 120$
 $\Delta U = 380 \text{ J}$
 In process AC
 $\Delta Q = \Delta U + \Delta W$
 $= 380 + \left[\frac{1}{2} \times (2 \times 10^4 + 6 \times 10^4) \times 2 \times 10^{-3} \right]$
 $= 380 + 80 = 460 \text{ J}$
[NEW NCERT Page No. 305]

15. (1)
 $\Delta U = -W$ for adiabatic process
[NEW NCERT Page No. 312]

16. (2)
 $\frac{dP}{dV} = -\frac{\gamma P}{V}$
[NEW NCERT Page No. 313]

17. (3)
 $PV = P' 0.9V \Rightarrow P' = \frac{10P}{9}$
[NEW NCERT Page No. 326]

18. (3)
 $\frac{V_2}{V_1} = \frac{T_2}{T_1} \Rightarrow \frac{T_2}{273} = 2 \Rightarrow T_2 = 546 \text{ K}$
[NEW NCERT Page No. 327]

19. (1)
 $v_{O_2} = \sqrt{\frac{3R \times 400}{32}} = \frac{1}{2} \sqrt{\frac{3R \times T}{2}}$
 $\Rightarrow \frac{T}{2} = 4 \times \frac{400}{32}$
 $\Rightarrow T = 100 \text{ K}$
[NEW NCERT Page No. 326]

20. (2)
 $v_{rms} = \sqrt{\frac{1^2 + 2^2 + 3^2 + 4^2 + 5^2}{5}} = \sqrt{11} \text{ km/s}$
 $v_{mean} = \frac{1 + 2 + 3 + 4 + 5}{5} = 3 \text{ km/s}$
[NEW NCERT Page No. 332]

21. (3)
 $P = \frac{nRT}{V} = \frac{2RT}{V}$
 $P' = \frac{4R2T}{V} = 4P$
[NEW NCERT Page No. 334]

22. (4)
 $N_1 C_v T_1 + N_2 C_v T_2 = (N_1 + N_2) C_v \cdot T_{mix}$
 $\therefore T_{mix} = \frac{N_1 T_1 + N_2 T_2}{N_1 + N_2}$
[NEW NCERT Page No. 335]

23. (2)
 $P = \frac{1}{3} M n V^2$
 $P' = \frac{1}{3} \times \frac{M}{2} n (2V)^2 = 2 \times \frac{1}{3} M n V^2 = 2P$
[NEW NCERT Page No. 334]

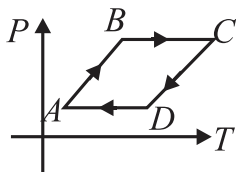
24. (3)
 For perfect gas,
 Kinetic Energy of n molecule, K.E
 $= n \left(\frac{1}{2} K_B T \right)$
 Where, K_B is Boltzmann constant
 If there is no loss of energy.
 Total kinetic energy of mixture is sum of each gas kinetic energy.
 $n_{total} K.E_{total} = n_1 K.E_1 + n_2 K.E_2 + n_3 K.E_3$
 $(n_1 + n_2 + n_3) \left(\frac{1}{2} K_B T \right) = n_1 \left(\frac{1}{2} K_B T_1 \right) + n_2 \left(\frac{1}{2} K_B T_2 \right)$
 $T = \frac{n_1 T_1 + n_2 T_2 + n_3 T_3}{n_1 + n_2 + n_3}$
[NEW NCERT Page No. 334]

25. (4)
 Both He and O₂ have 3 translatory degree of freedom. At the same temperature, energy is divided equally in all degrees of freedom.
 Hence ratio of the translatory kinetic energy is one.
[NEW NCERT Page No. 336]

26. (2)
 For cyclic forces $\Delta U = 0$ So, $\Delta Q = \Delta W$
[NEW NCERT Page No. 313]

27. (2)
 $\lambda_m T = \text{constant}$
 $\lambda_m \propto \frac{1}{T}$
 $\lambda_2 > \lambda_3 > \lambda_1$
[NEW NCERT Page No. 295]

28. (1)



$$v = \text{const.}$$

$$P \propto T$$

$$v = T$$

$$v = \text{constant}$$

$$p \propto T$$

$$P = \text{constant}$$

$$v \propto T$$

[NEW NCERT Page No. 316]

29. (3)

$$pV = nRT$$

$$T = \frac{v}{nR} P$$

$$\tan \theta = m = v$$

$$\theta_1 < \theta_2 < \theta_3$$

$$\tan \theta_1 < \tan \theta_2 < \tan \theta_3$$

$$V_1 < V_2 < V_3$$

[NEW NCERT Page No. 289]

30. (3)

Heat energy required to bring water from 24°C to 90°C is:

$$H = m.s.\Delta T = 100 \times 1 \times (90 - 24) = 6600 \text{ cal}$$

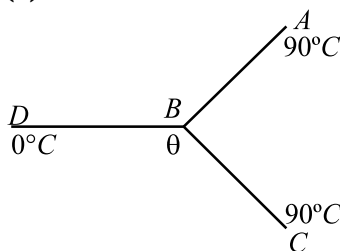
Amount of steam required for this heat energy is:

$$6600 = m(540) \quad [L_s = 540 \text{ cal/g}]$$

$$m = 12.22 \text{ g}$$

[NEW NCERT Page No. 290]

31. (2)



Rate of heat flow

$$(H) = \frac{\text{Temperature difference}(\Delta\theta)}{\text{Thermal resistance}(R)}$$

$$\text{where } R = \frac{L}{KA}$$

K = Thermal conductivity of the rod.

L = Length of rod.

A = Area of cross-section.

Let θ be the temperature of the junction (say B).

Thermal resistance (R) of all the three rods is equal.

Rate of heat flow through AB + Rate of heat flow through CB = Rate of heat flow through BD.

$$\therefore \frac{90 - \theta}{R} + \frac{90 - \theta}{R} = \frac{\theta - 0}{R}$$

$$\therefore 3\theta = 180^\circ\text{C}$$

$$\text{or } \theta = 60^\circ\text{C}$$

[New NCERT 11th, Part-II Page No. 288]

32. (1)

Temperature on Celsius scale (T_C), kelvin scale (T_K) and Fahrenheit scale (T_F) are related as

$$\frac{T_C - 0}{100} = \frac{T_F - 32}{180} = \frac{T_K - 273.15}{100}$$

[New NCERT 11th, Part-II Page No. 275]

33. (4)

Heat content of a body depends on Mass of the body, Temperature of the body and specific heat capacity.

[New NCERT 11th, Part-II Page No. 285]

34. (2)

The volume of liquid displaced by floating ice

$$V_D = \frac{M}{\sigma_L}$$

Volume of water formed by melting ice,

$$V_F = \frac{M}{\sigma_W}$$

$$\text{If } \sigma_L > \sigma_W, \text{ then } \frac{M}{\sigma_L} < \frac{M}{\sigma_W} \text{ i.e., } V_D < V_F$$

i.e., volume of liquid displaced by floating ice will be lesser than water formed and so the level of the liquid will rise.

[New NCERT 11th, Part-II Page No. 248]

35. (1)

Heat is carried away from a fire sideways mainly by radiations. Above the fire, heat is carried by both radiation and by convection of air. The latter process carries much more heat. So, assertion is true and reason is false.



[NCERT Page No. 292]

36. (3)

From Wein's law

$$\left[\lambda \propto \frac{1}{T} \right]$$

[New NCERT Class 11th Page No. 219]

37. (2)
 $(ms) = \text{thermal capacity}$
 $\frac{C_1}{C_2} = \frac{m_1 s}{m_2 s}$ ($s = \text{same for same substance}$)
 $\frac{C_1}{C_2} = \frac{R_1^3}{R_2^3} \Rightarrow \frac{C_1}{C_2} = \frac{1}{8}$
[NCERT Class 11th, Page No. 208]

38. (4)
 In parallel $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$
 $\frac{K_{eq}(2A)}{\ell} = \frac{K_1 A}{\ell} + \frac{K_2 A}{\ell}$
 $K_{eq} = \frac{K_1 + K_2}{2}$
[NCERT Class 11th, Page No. 215, 216]

39. (4)
 $Q = C\Delta T$
 $\Rightarrow \frac{dQ}{dt} = C \frac{dT}{dt}$
 Here $C = \text{thermal capacity}$.
 $\frac{(dT/dt)_1}{(dT/dt)_2} = \frac{C_2}{C_1} = \frac{8}{1}$
[NCERT Class 11th Page No. 219, 220]

40. (2)
 $\frac{E_1}{E_2} = \frac{\sigma T_1^4}{\sigma T_2^4}$
 $\Rightarrow \frac{E}{2E} = \left(\frac{1000}{T}\right)^4$
 $T = 2^{1/4} \times 1000 = (\sqrt{2})^{1/2} \times 1000$
 $1.19 \times 1000 = 1190K$
[NCERT Class 11th Page No. 219]

41. (2)
 $\frac{Q}{Q_2} = \frac{T^4}{(0.25)(2T)^4}$
 $\frac{Q}{Q_2} = \frac{1}{4}$
 $Q_2 = 4Q$
[NCERT Class 11th Page No. 219]

42. (3)
 At 4 °C water density is maximum. So, density first increase and then decrease.
[New NCERT Class 11th Page No. 207]

43. (3)
 On heating the system; x, r, d all increases, since the expansion of isotropic solids is similar to true photographic enlargement.
[New NCERT Class 11th Page No. 208]

44. (3)
 Specific gravity of alloy = $\frac{\text{Density of alloy}}{\text{Density of water}}$
 $= \frac{\text{Mass of alloy}}{\text{Volume of alloy} \times \text{density of water}}$
 $= \frac{m_1 + m_2}{\left(\frac{m_1}{\rho_1} + \frac{m_2}{\rho_2}\right) \times \rho_w} = \frac{m_1 + m_2}{\frac{m_1}{\rho_1 / \rho_w} + \frac{m_2}{\rho_2 / \rho_w}} = \frac{m_1 + m_2}{\frac{m_1}{s_1} + \frac{m_2}{s_2}}$
 [As specific gravity of substance = $\frac{\text{density of substance}}{\text{density of water}}$]
[New NCERT Class 11th Page No. 189]

45. (3)
 $\frac{39 - 0}{100 - 0} = \frac{T - 39}{239 - 39}$
 $117^\circ W = T$
[New NCERT Class 11th Page No. 275]

46. (1)
 Pressure at the bottom of tank
 $P = h\rho g = 3 \times 10^5 \frac{N}{m^2}$
 Pressure due to liquid column
 $P_l = 3 \times 10^5 - 1 \times 10^5 = 2 \times 10^5$
 and velocity of water $v = \sqrt{2gh}$
 $\therefore v = \sqrt{\frac{2P_l}{\rho}} = \sqrt{\frac{2 \times 2 \times 10^5}{10^3}} = \sqrt{400} \text{ m/s}$
[New NCERT Class 11th Page No. 332]

47. (1)
 If temperature in Celsius is t then
 $t = \frac{5}{9}(5t - 32)$
 $9t = 25t - 160$
 $16t = 160$
 $t = 10^\circ C$
[New NCERT Class 11th Page No. 275]

48. (1)
 $\ell_2' - \ell_1' = \ell_2 - \ell_1$
 $\Rightarrow \ell_2(1 + \alpha_2 \Delta t) - \ell_1(1 + \alpha_1 \Delta t) = \ell_2 - \ell_1$
 $\ell_2 \alpha_2 = \ell_1 \alpha_1$
[New NCERT Class 11th Page No. 170]

49. (4)
 $\frac{C}{5} = \frac{F - 32}{9} \Rightarrow \frac{C}{5} = \frac{140 - 32}{9} \Rightarrow C = 60^\circ C$
 $C = 60^\circ C$
[New NCERT Class 11th Page No. 275]

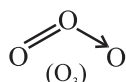
50. (2)
 $\theta_{\text{mix}} = \frac{m_w \theta_w - \frac{m_i L_i}{c_w}}{m_i + m_w}$
 For water and ice mixing
 $= \frac{20 \times 40 - \frac{5 \times 80}{1}}{5 + 20} = 16^\circ C$
[New NCERT Class 11th Page No. 334]

SECTION-(II) CHEMISTRY

51. (1)

Coordinate bond is a type of bond which is formed when a pair of electron is contributed by one atom and shared by both the atoms.

In case of O_3 , the central oxygen forms covalent bond with one oxygen, and coordinate bond with other oxygen as to complete its octet. The structure is shown below:

[New NCERT Class 11th Page No. 106]

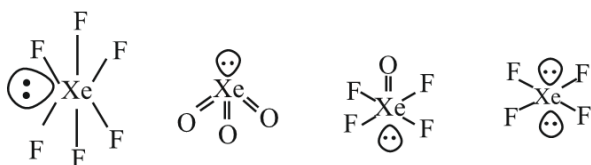
52. (1)

$XeF_6 \Rightarrow$ Distorted octahedral

$XeO_3 \Rightarrow$ Pyramidal

$XeOF_4 \Rightarrow$ Square pyramidal

$XeF_4 \Rightarrow$ Square planar

[New NCERT Class 11th Page No. 112]

53. (3)

Geometry of NH_3 molecule is tetrahedral and hybridization of 'N' in NH_3 is sp^3 .

[New NCERT Class 11th Page No. 120]

54. (2)

More the number of single bonds, more will be X–O bond length. There is no double bond present in the structure of SiO_4^{4-} , there is one double bond present in the structure of PO_4^{3-} , there are two double bonds present in the structure of SO_4^{2-} , and there are three double bonds present in the structure of ClO_4^- . Thus, the order of X–O bond length will be, $SiO_4^{4-} > PO_4^{3-} > SO_4^{2-} > ClO_4^-$

[New NCERT Class 11th Page No. 109]

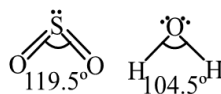
55. (4)

A π -bond is formed by the overlap of p-p orbitals in sideways manner.

[New NCERT Class 11th Page No. 119]

56. (1)

The shape of both SO_2 and H_2O are bent or V-shaped. These are shown as,



Bond angle of SO_2 is greater than that of H_2O due to higher repulsion of multiple bonds.

[New NCERT Class 11th Page No. 108]

57. (1)

The formal charge on water molecule is



$$F.C. = V.E. - N.E. - (B.E./2)$$

$$\text{Formula charge on H-atom} = 1 - 0 - (2/2) = 0$$

where, F.C = formal charge

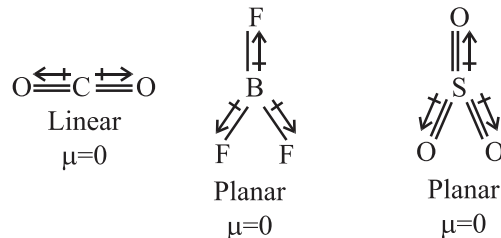
V.E. = number of valence electrons

N.E. = number of non-bonding electrons

B.E. = number of bond pair electrons

[New NCERT Class 11th Page No. 104]

58. (4)



All bond moments cancel out each other so dipole moment (μ) is zero for all of these.

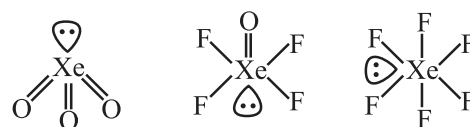
[New NCERT Class 11th Page No. 111]

59. (1)

According to Fajan's rule, covalent character in ionic compounds is directly proportional to polarizability of anion and polarizing power of cation. So, cation should be small and anion should be large in size for greater polarization.

[New NCERT Class 11th Page No. 106]

60. (4)

[New NCERT Class 11th Page No. 112]

61. (3)
 $sp^3 \quad sp^2 \quad sp \quad sp^2 \quad sp^3$
 $CH_3 - CH = C = CH - CH_3$
[New NCERT Class 11th Page No. 120]

62. (4)
 All three molecules, $AlCl_3$, SF_6 and NO , deviate from the octet rule, as the central atom in them does not contain exactly 8 electrons.
 The central atom in $AlCl_3$ has 6 valence electrons.
 The central atom in SF_6 has 12 valence electrons.
 The central atom in NO has 7 valence electrons.
[New NCERT Class 11th Page No. 115]

63. (4)
 NH_2^- , ClF_3 and H_2S have 2 lone pairs of electrons around the central atom. BF_3 has 3 bond pairs and zero lone pair of electrons around boron atom.
[New NCERT Class 11th Page No. 115]

64. (3)
 According to Fajan's rule large size (Rb^+) cation will have low polarisation power so have less covalent character and high ionic character.
[New NCERT Class 11th Page No. 106]

65. (1)
 Lattice energy $\propto \frac{1}{\text{size of anion}}$
[New NCERT Class 11th Page No. 107]

66. (3)
 $2p-2p > 2s-2p > 2s-2s$
[New NCERT Class 11th Page No. 119]

67. (2)
 Bond order for $O_2 = 2$ and for $O_2^+ = 2.5$
 Both are paramagnetic due to the presence of unpaired electrons (O_2 has 2 unpaired electron, O_2^+ has one unpaired electron).
[New NCERT Class 11th Page No. 129]

68. (2)
 Hybridization of $NO_3^- = \frac{1}{2}(5 + 0 + 1 - 0)$

$$= \frac{6}{2} = 3 = sp^2$$

 Hence geometry is trigonal planar.
[New NCERT Class 11th Page No. 122]

69. (2)
 By drawing the Lewis structures of these molecules.
 $SO_2 - sp^2$ and it will form two sigma bonds and 2π bonds. S will form $1(p_\pi - p_\pi)$ and $1(p_\pi - d_\pi)$ bonds.
 $SO_3 - sp^2$ and it will form three sigma bonds and 3π bonds. S will form $1(p_\pi - p_\pi)$ and $(2p_\pi - d_\pi)$ bonds.
 $ClO_4^- - sp^3$ and it will form four sigma bonds and 3π bonds. S will form $0(p_\pi - p_\pi)$ and $(3p_\pi - d_\pi)$ bonds.
 So, 1, 2 and 3 are correct answers option (2).
[New NCERT Class 11th Page No. 122]

70. (2)
- | Species | Hybridisation of 'Al' |
|------------|-----------------------|
| $AlCl_3$ | sp^2 |
| $AlCl_4^-$ | sp^3 |
- [New NCERT Class 11th Page No. 122]**

71. (3)
 KCN has ionic linkage between K^+ and CN^- .
 It has covalent linkage between C and N i.e. $C \equiv N^-$.
[New NCERT Class 11th Page No. 106]

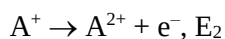
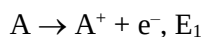
72. (1)
 ClF_3 has two lone pairs arranged at 120° to each other.
[New NCERT Class 11th Page No. 114]

73. (3)
 The vapour pressure of o-nitro phenol is greater than p-nitro phenol. It is due to intermolecular H-bonding in p-nitrophenol.
[New NCERT Class 11th Page No. 132]

74. (2)
 Sigma bond is formed by the end to end (head on) overlap of bonding orbitals along the internuclear axis. If the internuclear axis is z-axis, orbitals p_z and p_z overlap with each other to form σ bond.
[New NCERT Class 11th Page No. 120]

75. (1)
 $(\pi_{2p_x} = \pi_{2p_y}) < \sigma_{2p_z} < (\pi_{2p_y}^* = \pi_{2p_x}^*) < \sigma_{2p_z}^*$
[New NCERT Class 11th Page No. 129]

76. (1)



E_1 represents first I.E. of A and E_2 represent second I.E of A.

Since, second I.E. > First I.E.

So, $E_2 > E_1$

[New NCERT Class 11th Page No. 87]

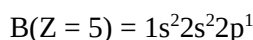
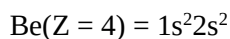
77. (1)

The high electron density around the small fluorine nucleus makes the electron affinity of fluorine lower than that of chlorine. Due to small size incoming electron feels more electron-electron repulsion so addition of electron is difficult as compared to Cl.

[New NCERT Class 11th Page No. 89]

78. (4)

First ionisation potential of Be (completely filled orbitals) is greater than boron due to stable configuration

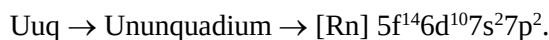


Order of attraction of electrons towards nucleus

$2s > 2p$, so more amount of energy is required to remove the electrons with 2s orbitals in comparison to 2p orbitals. So ionisation potential of Be is 9.32 eV and B is 8.29 eV.

[New NCERT Class 11th Page No. 87]

79. (1)



It belongs to carbon family.

[New NCERT Class 11th Page No. 80]

80. (3)

Electron gain enthalpy generally becomes more negative across a period as we move from left to right. Within a group, electron gain enthalpy becomes less negative down a group. However, adding an electron to the 2p-orbital leads to greater repulsion than adding an electron to the larger 3p-orbital, hence the element with most negative electron gain enthalpy is chlorine: the one with the least negative electron gain enthalpy is phosphorus.

[New NCERT Class 11th Page No. 89]

81. (1)

$P > Cl$ (Atomic size)

Atomic size decreases from left to right

Hence, size of $Cl < P$

[New NCERT Class 11th Page No. 86]

82. (2)

Ionization energy second of Na is higher than that of Mg because in case of Na, the second e^- has to be removed from the stable noble gas configuration while in case of Mg removal of second e^- gives a more stable noble gas configuration. Mg has high first ionisation enthalpy than Na because of its stable $3s^2$ configuration.

[New NCERT Class 11th Page No. 87]

83. (3)

Noble gases have positive electron gain enthalpy as they have fully filled electronic configuration and hence it is difficult to add an electron to it.

[New NCERT Class 11th Page No. 89]

84. (3)

Hydrogen is a non metal and non metals are more electronegative than metals.

[New NCERT Class 11th Page No. 90]

85. (4)

Screening effect is not observed in monoelectronic species.

$\text{He}^+, \text{Li}^{2+}, \text{Be}^{3+} \Rightarrow$ All are monoelectronic species.

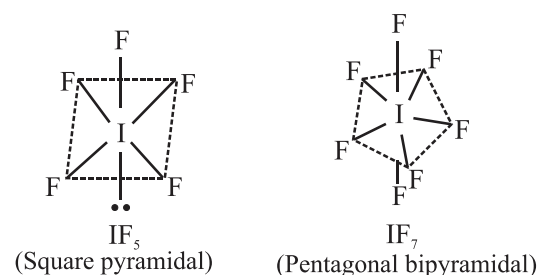
[New NCERT Class 11th Page No. 89]

86. (2)

$\text{Cd}(Z = 48) : [\text{Kr}] 4d^{10} 5s^2 \Rightarrow$ 5th period, 12th group

[New NCERT Class 11th Page No. 81]

87. (1)



[New NCERT Class 11th Page No. 113]

88. (3)

Total number of electron in N_2 are 14, so bond order is 3 and it is diamagnetic in nature.

[New NCERT Class 11th Page No. 129]

89. (4)

$Li < Na < K$ (correct order of atomic size)

$Li > Be > B$ (correct order of atomic size)

Ionic size $\propto \frac{-ve \text{ charge}}{+ve \text{ charge}}$ (For isoelectronic species)

[New NCERT Class 11th Page No. 86]

90. (2)

Metallic nature decreases, acidic nature increases.

[New NCERT Class 11th Page No. 85]

91. (2)

- Arsenic is a p-block element.
- Arsenic belongs to group-15.

[New NCERT Class 11th Page No. 79]

92. (4)

- Chlorine has Highest negative electron gain enthalpy.
- Cesium is Most electropositive element.
- Lithium is stronger reducing agent.
- Helium has Highest ionisation enthalpy.

[New NCERT Class 11th Page No. 89]

93. (2)

- The atomic radius of gallium is not higher than that of aluminum. In fact, gallium has a slightly smaller atomic radius compared to aluminum, despite being in the same group, due to the poor shielding effect of the 3d electrons.
- The presence of additional 10 d-electrons in gallium indeed provides a poor screening effect for the outer electrons from the increased nuclear charge. This poor shielding effect causes the outer electrons to be more strongly attracted to the nucleus, effectively reducing the atomic radius.

[New NCERT Class 11th Page No. 85]

94. (1)

The size of isoelectronic species F^- and Na^+ is affected by effective nuclear charge (Z_{eff}).

effective nuclear charge $\propto$ Charge on cation and

effective nuclear charge $\propto \frac{1}{\text{Charge on Anion}}$

[New NCERT Class 11th Page No. 85]

95. (4)

Atomic radii are:

$C = 77 \text{ pm}$, $N = 74 \text{ pm}$, $O = 66 \text{ pm}$, $F = 65 \text{ pm}$

So correct order is $F < O < N < C$

[New NCERT Class 11th Page No. 92]

96. (1)

On taking x-axis as the inter nuclear axis, the highest occupied MO (or molecular orbital) in N_2 and O_2^+ are $\sigma 2p_x$ and $\pi^* 2p_y$, respectively.

[New NCERT Class 11th Page No. 129]

97. (2)

The f-block elements belong to Group-3 of the modern periodic table.

[New NCERT Class 11th Page No. 84]

98. (4)

Among isoelectronic monoatomic species, size is inversely proportional to atomic number. Hence among isoelectronic species Na^+ , O^{2-} , N^{3-} , F^- (having nearest noble gas configuration); Order of size is $Na^+ < F^- < O^{2-} < N^{3-}$

N^{3-} has least atomic number hence largest size.

[New NCERT Class 11th Page No. 79]

99. (1)

Bond order $\propto$ Bond dissociation energy

Higher is the bond order, greater is the bond dissociation energy.

[New NCERT Class 11th Page No. 129]

100. (1)

Oxygen atom possesses two lone pair of electrons. Due to lone pair-lone pair repulsion, bond angle in H_2O decreases from $109^\circ 28'$ to 104.5° .

[New NCERT Class 11th Page No. 115]

SECTION-(III) BOTANY**101. (4)**

In Malvaceae family, pentamerous flowers with epicalyx, corolla with mucilage, androecium with numerous stamens, pentacarpellary, syncarpous, superior ovary and axile placentation.

[New NCERT Class 11th Page No. 64, 65]

102. (1)

Placentation in Fabaceae family is marginal.

[New NCERT Class 11th Page No. 65]

103. (3)

Bryophytes are also called amphibians of the plant kingdom because these plants can live in soil but are dependent on water for sexual reproduction.

[New NCERT Class 11th Page No. 29]

104. (1)

In pheophyceae, the plant body is usually attached to the substratum by a holdfast, and has a stalk, the stipe and leaf like photosynthetic organ-the frond.

[New NCERT Class 11th Page No. 27]

105. (1)

In Asteraceae family, flowers are actinomorphic (disc florets) or zygomorphic (ray florets) and monocarpellary, syncarpous and inferior ovary. Ray florets are found towards the periphery (they are older) and disc florets are distributed near the center (thus younger).

[New NCERT Class 11th Page No. 62]

106. (1)

In red algae, food is stored as floridean starch which is very similar to amylopectin and glycogen in structure. Example; *Polysiphonia*, *Porphyra*, *Gracilaria* and *Gelidium*.

[New NCERT Class 11th Page No. 27]

107. (2)

From the region of maturation some of the epidermal cells form very fine and delicate, thread-like structures called root hairs.

[New NCERT Class 11th Page No. 59]

108. (3)

In pteridophytes, the spores germinate to give rise to inconspicuous, small but multicellular, free-living, mostly photosynthetic thalloid gametophytes called prothallus.

[New NCERT Class 11th Page No. 32]

109. (3)

Brassicaceae	Mustard
Poaceae	Wheat
Asteraceae	Marigold
Malvaceae	China rose

[Old NCERT Class 11th Page No. 69]

110. (1)

Offsets are found in *Pistia*.

[Old NCERT Class 11th Page No. 69]

111. (2)

Most of the members of chlorophyceae have one or more storage bodies called pyrenoids located in the chloroplasts. Pyrenoids contain protein besides starch.

[New NCERT Class 11th Page No. 26]

112. (3)

Inflorescence in the members of family Brassicaceae is said to be racemose in which floral axis continues to grow and the flowers are borne laterally in an acropetal succession.

[New NCERT Class 11th Page No. 61]

113. (1)

In liverworts, spores produced on meiosis germinate to form free-living gametophytes.

[New NCERT Class 11th Page No. 29]

114. (1)

Bryophytes in general are of little economic importance but some mosses provide food for herbaceous mammals birds and other animals.

[New NCERT Class 11th Page No. 29]

115. (1)

The plumule and radicle are enclosed in sheaths which are called coleoptile and coleorhiza respectively.

[New NCERT Class 11th Page No. 67]

116. (4)

Phycocyanin, chlorophyll *a* and chlorophyll *b* are characteristics of Rhodophyceae.

[New NCERT Class 11th Page No. 27]

117. (3)

Spine of *Aloe* are modification of leaves that protect the plant from predators.

[Old NCERT Class 11th Page No. 71]

118. (3)
In palmately compound leaves, the leaflets are attached at a common point, i.e., at the tip of petiole, as in silk cotton.

[New NCERT Class 11th Page No. 61]

119. (1)
In bryophytes, the sporophyte is not free-living but attached to the photosynthetic gametophyte and derives nourishment from it.

[New NCERT Class 11th Page No. 29]

120. (2)
Diadelphous-Pea

[New NCERT Class 11th Page No. 64]

121. (3)
Each anther is usually bilobed and each lobe has two chambers, the pollen-sacs. The pollen grains are produced in pollen-sacs. A sterile stamen is called staminode.

[New NCERT Class 11th Page No.64]

122. (3)
In *Cycas*, small specialised roots called coralloid roots are associated with N₂-fixing cyanobacteria.

[New NCERT Class 11th Page No. 32]

123. (4)
In epigynous flowers, the margin of thalamus grows upward enclosing the ovary completely and getting fused with it, the other parts of flower arise above the ovary. Hence, the ovary is said to be inferior as in flowers of guava and cucumber, and the ray florets of sunflower.

[New NCERT Class 11th Page No. 63]

124. (4)
Funaria possess multicellular and branched rhizoids.

[New NCERT Class 11th Page No. 30]

125 (2)
If the margins of sepals or petals overlap one another but not in any particular direction as in *Cassia* and gulmohar, the aestivation is called imbricate.

[New NCERT Class 11th Page No. 64]

126. (3)
Genera like *Selaginella* and *Salvinia* which produce two kinds of spores, macro (large) and micro (small) spores, are known as heterosporous.

[New NCERT Class 11th Page No. 32]

127. (1)
In mango and coconut, the fruit is known as a drupe). They develop from monocarpellary superior ovaries and are one seeded.

[New NCERT Class 11th Page No.65]

128. (2)

Psilopsida	<i>Psilotum</i>
Lycopsida	<i>Selaginella</i>
Sphenopsida	<i>Equisetum</i>
Pteropsida	<i>Dryopteris</i>

[New NCERT Class 11th Page No. 32]

129. (2)
⊕ represents actinomorphic flowers and % represents zygomorphic flower.

[New NCERT Class 11th Page No. 67]

130. (4)
The stems are unbranched (*Cycas*) or branched (*Pinus*, *Cedrus*).

[New NCERT Class 11th Page No. 32]

131. (1)
In an alternate type of phyllotaxy, a single leaf arises at each node in an alternate manner, as in china rose, mustard and sunflower plants.

[New NCERT Class 11th Page No. 61]

132. (3)
Scutellum – Cotyledons of Monocot seeds.

[Old NCERT Class 11th Page No. 77]

133. (3)
In the members of Poaceae family the ovary is monocarpellary, unilocular and superior.

[New NCERT Class 11th Page No. 65]

134. (1)
The sporophyte is differentiated into a foot, seta and capsule in liverworts and mosses.

[New NCERT Class 11th Page No. 30]

135. (3)
The female gametophytes in pteridophytes retained on the parent sporophytes for variable periods. The development of the zygotes into young embryos take place within the female gametophytes. This event is a precursor to the seed habit considered an important step in evolution.

[New NCERT Class 11th Page No. 32]

136. (4)

Leaves originate from shoot apical meristems and are arranged in an acropetal order.

[New NCERT Class 11th Page No. 59]

137. (2)

In gymnosperm, the microspores develop into a male gametophytic generation which is highly reduced and is confined to only a limited number of cells. This reduced gametophyte is called a pollen grain.

[New NCERT Class 11th Page No. 33]

138. (2)

When a flower can be divided into two equal radial halves in any radial plane passing through the centre, it is said to be actinomorphic, e.g., mustard, datura, chilli.

[New NCERT Class 11th Page No. 62]

139. (1)

After fertilisation, the ovary is modified into fruits and ovules into seeds.

[New NCERT Class 11th Page No. 65,66]

140. (3)

The leaf is attached to the stem by the leaf base and may bear two lateral small leaf like structures called stipules.

[New NCERT Class 11th Page No. 60]

141. (3)

The gymnosperms are heterosporous; they produce haploid microspores and megaspores. *Salvinia* is a pteridophyte. The leaves in gymnosperms are well-adapted to withstand extremes of temperature, humidity and wind.

[New NCERT Class 11th Page No. 33]

142. (3)

When stamens are attached to the petals, they are epipetalous as in brinjal, or epiphyllous when attached to the perianth as in the flowers of lily.

[New NCERT Class 11th Page No. 64]

143. (2)

The given figure represents free central placentation.

[New NCERT Class 11th Page No. 65]

144. (1)

Phylogenetic classification systems based on evolutionary relationships between the various organisms.

[New NCERT Class 11th Page No. 24]

145. (1)

Fruits of brinjal plants have persistent calyx.

[Old NCERT Class 11th Page No. 80]

146. (4)

Fusion between one large, nonmotile (static) female gamete and a smaller, motile male gamete is termed oogamous, e.g., *Volvox fucus*.

[New NCERT Class 11th Page No. 24]

147. (4)

In mango the pericarp is well differentiated into an outer thin epicarp, a middle fleshy edible mesocarp and an inner stony hard endocarp.

[New NCERT Class 11th Page No. 66]

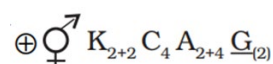
148. (1)

A typical flower has four different kinds of whorls arranged successively on the swollen end of the stalk or pedicel, called thalamus or receptacle.

[New NCERT Class 11th Page No. 62]

149. (1)

The correct floral formula for mustard plant (Family Brassicaceae) is



[New NCERT Class 11th Page No. 68]

150. (2)

Artificial classification separated the closely related species since they were based on a few characteristics.

[New NCERT Class 11th Page No. 23]

SECTION-(IV) ZOOLOGY**151. (2)**

The sarcomere is the smallest functional unit of a skeletal muscle fiber responsible for contraction

[New NCERT Class 11th Page No. 220]

152. (1)

The first seven rib pairs, called true ribs, attach dorsally to thoracic vertebrae and ventrally to the sternum via hyaline cartilage.

[New NCERT Class 11th Page No. 225]

153. (4)

The facial region is made up of 14 skeletal elements.

[New NCERT Class 11th Page No. 227]

154. (3)

Cardiac muscle is specialised for the heart, having unique properties such as involuntary contractions.

[New NCERT Class 11th Page No. 218]

155. (3)

The acromion is a bony projection that extends from the scapula.

[New NCERT Class 11th Page No. 221]

156. (1)

- Each vertebra has a central hollow portion (neural canal) through which the spinal cord passes.
- First vertebra is the atlas and it articulates with the occipital condyles

[New NCERT Class 11th Page No. 225]

157. (1)

The light bands contain actin and is called I-band or Isotropic band.

[New NCERT Class 11th Page No. 219]

158. (2)

- Joints are points of contact between bones, or between bones and cartilages.
- Force generated by the muscles is used to carry out movement through joints, where the joint acts as fulcrum.

[New NCERT Class 11th Page No. 218]

159. (1)

Acetylcholinesterase is an enzyme that breaks down acetylcholine at the neuromuscular junction, leading to muscle relaxation by stopping continuous stimulation of the muscle.

[New NCERT Class 11th Page No. 22]

160. (4)

Accumulation of uric acid crystals occurs in gout disease.

[New NCERT Class 11th Page No. 227]

161. (1)

- Muscular dystrophy is a genetic disorder.
- Scapula is a large triangular flat bone situated in the dorsal part of the thorax between the second and the seventh ribs.

[New NCERT Class 11th Page No. 226]

162. (3)

The joint between the carpal and the metacarpal of thumb is a saddle joint.

[New NCERT Class 11th Page No. 224]

163. (3)

- The central part of thick filament, not overlapped by thin filaments is called the 'H' zone.
- The thick filaments in the 'A' band are held together in the middle of the band by a thin fibrous membrane called 'M' line.

[New NCERT Class 11th Page No. 222]

164. (4)

Synovial fluid lubricates joints, absorbs shock, and provides nutrients to cartilage.

[New NCERT Class 11th Page No. 227]

165. (4)

Metatarsals (5 in number) while tarsals (ankle bones – 7 in number).

[New NCERT Class 11th Page No. 226]

166. (1)

A subunit of troponin blocks the binding sites in the absence of calcium, preventing contraction.

[New NCERT Class 11th Page No. 221]

167. (1)

The U-shaped bone at the base of the buccal cavity is hyoid.

[New NCERT Class 11th Page no. 224]

168. (2)

Myoglobin binds oxygen and facilitates its storage and availability in muscle cells.

[New NCERT Class 11th Page No. 223]

169. (4)

The number of mitochondria are few in white muscle fibres.

[New NCERT Class 11th Page No. 223]

170. (4)
Macrophages and leucocytes exhibit amoeboid movement.

[New NCERT Class 11th Page No. 218]

171. (2)
Ball and socket joint is present between humerus and pectoral girdle.

[New NCERT Class 11th Page No. 227]

172. (4)
Both cardiac and skeletal muscles are striated.

[New NCERT Class 11th Page No. 218]

173. (2)
- Calcium ions are pumped back into the sarcoplasmic reticulum.
 - Explanation: Pumping calcium ions back into the sarcoplasmic reticulum leads to muscle relaxation.

[New NCERT Class 11th Page No. 222]

174. (3)
The sliding filament theory explains muscle contraction, where actin slides over myosin, resulting in shortening of the muscle fibers.

[New NCERT Class 11th Page No. 221]

175. (1)

Ciliary movement	Female reproductive tract
Myosin	ATP binding sites
Myoglobin	Red muscle fibers
Arthritis	Inflammation of joints

[New NCERT Class 11th Page No. 218, 221, 227]

176. (3)
Skeletal muscles are attached to bones and are responsible for voluntary movements.

[New NCERT Class 11th Page No. 218]

177. (1)
- The vertebral column consists of individual vertebrae, and intervertebral discs, made of fibrocartilage, provide cushioning and allow flexibility in the spine.
 - The number of cervical vertebrae are seven in almost all mammals including human beings.

[New NCERT Class 11th Page No. 225]

178. (4)
Visceral muscles, located in hollow organs like the alimentary canal, are smooth, non-striated, and involuntary.

[New NCERT Class 11th Page No. 218]

179. (1)

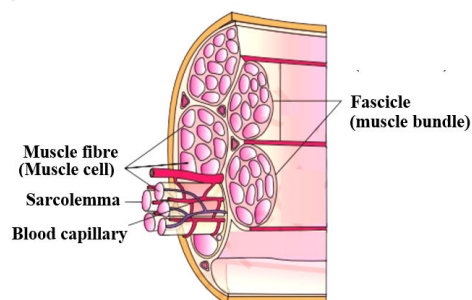


Figure: Diagrammatic cross sectional view of a muscle showing muscle bundles and muscle fibres.

[New NCERT Class 11th Page No. 219]

180. (1)
Rapid spasms in muscle due to low Ca^{++} in body fluid.

[New NCERT Class 11th Page No. 222]

181. (3)

Pseudopodia	Amoeboid movement
Actin	Thin filament
Cranial bones	Amoeboid movement
Femur	Longest bone

[New NCERT Class 11th Page No. 224]

182. (4)
The vertebral column protects the spinal cord, supports the head and serves as the point of attachment for the ribs and musculature of the back.

[New NCERT Class 11th Page no. 225]

183. (4)
Humerus is found in pectoral girdle.

[New NCERT Class 11th Page no. 226]

184. (3)
- HMM = Head + Short arm
 - LMM = Tail

[New NCERT Class 11th Page no. 221]

185. (4)
- Muscle contains a red-coloured oxygen storing pigment called myoglobin.
 - Amount of sarcoplasmic reticulum is high in white muscle fibres.
 - Patella is a cup shaped bone.
 - A motor neuron alongwith the muscle fibres connected to it constitute a motor unit.

[New NCERT Class 11th Page no. 223]

186. (4)

- Skeletal muscles are closely associated with the skeletal components of the body.
- They have a striped appearance under the microscope and hence are called striated muscles.
- As their activities are under the voluntary control of the nervous system, they are known as voluntary muscles too.
- They are primarily involved in locomotory actions and changes of body postures.

[New NCERT Class 11th Page no. 218]

187. (1)

Stimulation of a muscle fibre by a motor neuron occurs at the neuromuscular junction.

[New NCERT Class 11th Page no. 222]

188. (2)

- Muscle contraction is initiated by a signal sent by the central nervous system.
- The junction between a motor neuron and the sarcolemma of the muscle fibre is called the motor-end plate.

[New NCERT Class 11th Page no. 222]

189. (1)

Sarcoplasmic reticulum of skeletal muscle fibres is a storehouse of Ca^{2+} .

[New NCERT Class 11th Page no. 303]

190. (3)

- Dust particle inhaled along with the atmospheric air can be removed from trachea.
- Trachea is lined with ciliated epithelium.

[New NCERT Class 11th Page no. 218]

191. (4)

Middle ear contains three tiny bones called stapes, malleus and incus.

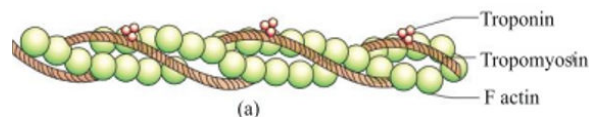
[New NCERT Class 11th Page no. 224]

192. (1)

Acetabulum is a cavity to which thigh bone articulates.

[New NCERT Class 11th Page no. 226]

193. (2)



[New NCERT Class 11th Page No. 221]

194. (2)

- Upper arm: Striated and voluntary
- Heart: Striated and involuntary
- Alimentary canal: Non-striated and involuntary

[New NCERT Class 11th Page no. 218]

195. (4)

Vertebral column is formed by 26 serially arranged units called vertebrae.

[New NCERT Class 11th Page no. 225]

196. (4)

Muscles with very low quantity of haemoglobin appear pale or whitish.

[New NCERT Class 11th Page no. 223]

197. (4)

Cranial bones	8
Forelimb	30
Ribs	24
Fibula	2

[New NCERT Class 11th Page no. 224, 225]

198. (1)

- Human skull is dicondylic.
- The two halves of the pelvic girdle meet ventrally to form the pubic symphysis containing fibrous cartilage.

[New NCERT Class 11th Page no. 224]

199. (1)

Muscle fibre is lined by the plasma membrane called sarcolemma.

[New NCERT Class 11th Page no. 219]

200. (4)

Each coxal bone is formed by the fusion of ilium, pubis and ischium.

[New NCERT Class 11th Page no. 226]

